ML Lab

Assignment 4:

a. **Applied various clustering algorithms using Python on the Iris and Wine datasets from UCI.**
b. **Implemented Partition-based clustering methods: K-Means, K-Means++, K-Medoids (PAM), and Bisecting K-Means.**
c. **Performed Hierarchical clustering using dendrograms with multiple linkage strategies (Ward, Average, Complete, Single).**
d. **Evaluated Density-based clustering methods: DBSCAN and OPTICS.**
e. **Compared performances using multiple metrics including Accuracy, Rand Index, Adjusted Rand Index, Mutual Information, Silhouette Coefficient, Calinski-Harabasz Index, Davies-Bouldin Index, and cohesion/separation scores (SSE & SSB).**
f. **Analyzed and identified the best-performing clustering approach for each dataset, achieving accuracy ≥80% for both Iris and Wine datasets.**

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Github Link:
https://github.com/abhijitdas6371/ML_LAB/blob/main/Ass4/Assignment4_ML_Lab.ipynb

1. **Problem Statement**

The goal is to implement and evaluate clustering algorithms on multiple datasets. The task is to group samples into clusters based on feature similarities, and compare algorithm performance using various internal and external metrics.

Datasets:

a. Iris dataset: Group iris plants into three species (Setosa, Versicolor, Virginica) based on four features (sepal length, sepal width, petal length, petal width).
b. Wine dataset: Cluster wine samples into three classes based on 13 chemical and physical features.

Clustering Algorithms Implemented:

a. Partition-based clustering: K-Means, K-Means++, K-Medoids (PAM), Bisecting K-Means
b. Hierarchical clustering: Dendrogram with multiple linkage strategies (Ward, Average, Complete, Single)
c. Density-based clustering: DBSCAN, OPTICS

Evaluation Metrics:

The performance of each algorithm was evaluated using both external and internal metrics:

I. **External metrics:** Rand Index, Adjusted Rand Index, Mutual Information, Adjusted Mutual Information, Normalized Mutual Information, Accuracy (via Hungarian mapping)
II. **Internal metrics:** Silhouette Coefficient, Calinski-Harabasz Index, Davies-Bouldin Index
III. **Cohesion and separation:** Sum of Squared Error (SSE) and Sum of Squares Between groups (SSB)

The experiments aimed to identify the best clustering approach for each dataset, with the goal of achieving accuracy ≥80% while analyzing cluster quality and separation.

2. **Discussion on different models**

K-Means Clustering

K-Means is a partition-based clustering algorithm that divides the dataset into k clusters by minimizing the sum of squared distances between data points and their respective cluster centroids.

Working Principle:

- Randomly initialize k centroids.
- Assign each data point to the nearest centroid.
- Update centroids as the mean of assigned points.
- Iterate until convergence (centroids do not change significantly).

Key Features:

- Simple and efficient for large datasets.
- Sensitive to initial centroid placement.
- Works best on spherical clusters with similar variance.

Parameters:

- n_clusters: Number of clusters to form.
- init: Method to initialize centroids (‘random’ or ‘k-means++’).
- n_init: Number of times the algorithm will run with different centroid seeds.
- max_iter: Maximum number of iterations per run.
- random_state: Controls random initialization for reproducibility.

K-Means++

K-Means++ is an enhanced version of K-Means that improves centroid initialization to speed up convergence and increase accuracy.

Working Principle:

- The first centroid is chosen randomly.
- Subsequent centroids are chosen with probability proportional to the squared distance from the nearest existing centroid.
- Standard K-Means steps are followed afterward.

Key Features:

- Reduces the likelihood of poor clustering due to random initialization.
- Often achieves better accuracy and faster convergence than random initialization.

Bisecting K-Means

Bisecting K-Means is a hierarchical variant of K-Means that repeatedly splits clusters to form the desired number of clusters.

Working Principle:

- Start with all points in a single cluster.
- Select the cluster with the largest SSE (Sum of Squared Errors).
- Split it into two using standard K-Means.
- Repeat until the desired number of clusters is achieved.

Key Features:

- Combines hierarchical and partition-based approaches.
- Often produces more balanced clusters than standard K-Means.

K-Medoids / PAM (Partitioning Around Medoids)

K-Medoids is similar to K-Means but chooses actual data points as cluster centers (medoids), making it more robust to outliers.

Working Principle:

- Initialize k medoids.
- Assign each point to the nearest medoid.
- Swap medoids with non-medoid points to reduce total distance.
- Repeat until convergence.

Key Features:

- More robust to noise and outliers than K-Means.
- Slightly slower due to combinatorial optimization.

Parameters:

- n_clusters: Number of clusters.

- method: PAM or alternative medoid selection algorithm.
- random_state: Seed for reproducibility.

Hierarchical Clustering (Dendrogram)

Hierarchical clustering builds a tree of clusters, called a dendrogram, without requiring a predefined number of clusters.

Working Principle:

- The agglomerative approach starts with each point as its own cluster and merges clusters iteratively based on distance.
- Linkage methods determine how distances between clusters are calculated (Ward, Average, Complete, Single).
- The dendrogram can be cut at a chosen height to form clusters.

Key Features:

- No need to specify k initially.
- Visualizes cluster relationships.
- Computationally intensive for large datasets.

DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

DBSCAN groups points based on density, identifying clusters as areas of high density separated by low-density regions.

Working Principle:

- Define a radius eps and minimum points min_samples.
- Points with at least min_samples neighbors within eps are core points.
- Expand clusters from core points; points not reachable from any core are noise.

Key Features:

- Detects arbitrarily shaped clusters.
- Handles outliers naturally.
- Does not require the number of clusters to be specified.

Parameters:

- eps: Neighborhood radius.
- min_samples: Minimum number of points to form a dense region.

OPTICS (Ordering Points To Identify the Clustering Structure)

OPTICS is a density-based clustering algorithm similar to DBSCAN but can identify clusters with varying densities.

Working Principle:

- Computes a reachability distance for each point.
- Orders points based on density-reachability.
- Clusters are extracted from the reachability plot without a fixed eps.

Key Features:

- Handles clusters of different densities.
- Naturally identifies noise.
- Requires post-processing of reachability distances to extract clusters.

Discussion on Iris Dataset

The Iris dataset contains 150 samples with 4 continuous features describing sepal and petal measurements for three species of Iris flowers. It is low-dimensional with partially overlapping clusters, making it suitable for partition-based, hierarchical, and density-based clustering approaches.

```
==== Dataset: Iris | samples=150 features=4 classes=3 ====
Running KMeans (best-of-N random init)...
 KMeans best accuracy: 0.8333
Running KMeans++ (scikit-learn with multiple n_init)...
 KMeans++ best accuracy: 0.8333
Running K-Medoids / PAM...
 KMedoids accuracy: 0.8467
Running Bisecting KMeans...
 Bisecting KMeans accuracy: 0.8333
Running Hierarchical clustering (linkages: ward, average, complete, single)...
  linkage=ward     acc=0.8267
  linkage=average  acc=0.6867
  linkage=complete acc=0.7867
  linkage=single   acc=0.6600
 Best hierarchical linkage: ward acc=0.8267
  Saved dendrogram to: plot_Iris_dendrogram_ward.png
Running DBSCAN (default eps/min_samples heuristics)...
 DBSCAN clusters (including noise=-1): [-1  0  1]
Running OPTICS...
 OPTICS clusters (including noise=-1): [-1  0  1  2  3  4]
```

=== Iris Results ===

dataset	algorithm	n_clusters	accuracy	rand_score	adjusted_rand_score	mutual_info	adjusted_mutual_info	normalized_mutual_info	silhouette	calinski_harabasz	davies_bouldin	SSE	SSB
Iris	KMeans (best-of-10 random)	3	0.8333	0.8322	0.6201	0.7241	0.6552	0.6595	0.4599	241.9044	0.8336	139.8205	460.1795
Iris	KMeans++ (best-of-10)	3	0.8333	0.8322	0.6201	0.7241	0.6552	0.6595	0.4599	241.9044	0.8336	139.8205	460.1795
Iris	KMedoids/PAM	3	0.8467	0.8415	0.6416	0.7405	0.6710	0.6750	0.4566	238.7630	0.8419	141.2271	458.7729
Iris	Bisecting KMeans	3	0.8333	0.8322	0.6201	0.7241	0.6552	0.6595	0.4599	241.9044	0.8336	139.8205	460.1795
Iris	Hierarchical (ward)	3	0.8267	0.8252	0.6153	0.7228	0.6713	0.6755	0.4467	222.7192	0.8035	148.8763	451.1237
Iris	DBSCAN (eps=0.5,min_samples=5)	3	0.6800	0.7476	0.4421	0.5499	0.5052	0.5114	0.3565	84.5103	7.1241	279.0957	320.9043
Iris	OPTICS (min_samples=5)	6	0.4467	0.5121	0.0514	0.3082	0.2657	0.2924	-0.3009	8.3282	2.4253	465.4143	134.5857

Partition-based Clustering

K-Means (best-of-10 random initialization)

Achieved accuracy: 0.833

Rand score: 0.832, Adjusted Rand score: 0.620

SSE: 139.82, SSB: 460.18

Interpretation: K-Means effectively separates Setosa from the other species but partially merges Versicolor and Virginica due to overlapping features. Multiple random initializations ensure more stable results.

K-Means++

Accuracy: 0.833 (same as K-Means)

Interpretation: Improved initialization does not significantly affect performance for this dataset but guarantees consistent convergence.

Bisecting K-Means

Accuracy: 0.833

Interpretation: Recursive splitting produces comparable clusters to standard K-Means, with similar cohesion and separation scores.

K-Medoids / PAM

Accuracy: 0.847 (best among partition-based methods)

Rand score: 0.842, Adjusted Rand: 0.642

SSE: 141.23, SSB: 458.77

Interpretation: Medoid-based clustering improves robustness to overlapping clusters and outliers, achieving slightly higher accuracy than K-Means.

Hierarchical Clustering (Dendrogram)

Best linkage: Ward, Accuracy: 0.827

Average, Complete, Single linkages performed worse (0.686–0.787)

Interpretation: Ward linkage minimizes intra-cluster variance, producing the most compact clusters. Dendrogram visualization confirms clear separation for Setosa and partial separation for the other two species.

Density-based Clustering

DBSCAN

Accuracy: 0.680, clusters identified: [-1, 0, 1]

Interpretation: DBSCAN correctly identifies Setosa but struggles with overlapping Versicolor and Virginica clusters. Some points near boundaries are treated as noise.

OPTICS

Accuracy: 0.447, clusters identified: [-1, 0, 1, 2, 3, 4]

Interpretation: Over-segmentation due to density variations reduces alignment with true class labels.

Conclusion

a. Best clustering method: K-Medoids / PAM (accuracy 0.847)
b. Hierarchical clustering with Ward linkage is also effective (accuracy 0.827)
c. Density-based methods perform less effectively due to uniform density and overlap.
d. Overall, partition-based clustering is most suitable for the Iris dataset.

Discussion on Wine Dataset

The Wine dataset contains 178 samples with 13 continuous features representing chemical properties of wines from three cultivars. It is higher-dimensional with correlated features, making it suitable for partition-based and hierarchical clustering.

```
==== Dataset: Wine | samples=178 features=13 classes=3 ====
Running KMeans (best-of-N random init)...
 KMeans best accuracy: 0.9719
Running KMeans++ (scikit-learn with multiple n_init)...
 KMeans++ best accuracy: 0.9663
Running K-Medoids / PAM...
 KMedoids accuracy: 0.9101
Running Bisecting KMeans...
 Bisecting KMeans accuracy: 0.8371
Running Hierarchical clustering (linkages: ward, average, complete, single)...
  linkage=ward     acc=0.9270
  linkage=average  acc=0.3876
  linkage=complete acc=0.8371
  linkage=single   acc=0.3764
 Best hierarchical linkage: ward acc=0.9270
  Saved dendrogram to: plot_Wine_dendrogram_ward.png
Running DBSCAN (default eps/min_samples heuristics)...
 DBSCAN clusters (including noise=-1): [-1]
Running OPTICS...
 OPTICS clusters (including noise=-1): [-1  0  1  2  3]
```

=== Wine Results ===

dataset	algorithm	n_clusters	accuracy	rand_score	adjusted_rand_score	mutual_info	adjusted_mutual_info	normalized_mutual_info	silhouette	calinski_harabasz	davies_bouldin	SSE	SSB
Wine	KMeans (best-of-10 random)	3	0.9719	0.9620	0.9149	0.9725	0.8914	0.8926	0.2859	70.8369	1.3918	1278.7608	1035.2392
Wine	KMeans++ (best-of-10)	3	0.9663	0.9543	0.8975	0.9545	0.8746	0.8759	0.2849	70.9400	1.3892	1277.9285	1036.0715
Wine	KMedoids/PAM	3	0.9101	0.8840	0.7411	0.8490	0.7806	0.7829	0.2676	67.1223	1.4248	1309.4810	1004.5190
Wine	Bisecting KMeans	3	0.8371	0.8130	0.5906	0.7515	0.7019	0.7051	0.2341	59.7425	1.5203	1375.1129	938.8871
Wine	Hierarchical (ward)	3	0.9270	0.9065	0.7899	0.8584	0.7842	0.7865	0.2774	67.6475	1.4186	1305.0487	1008.9513
Wine	DBSCAN (eps=0.5,min_samples=5)	1	0.3989	0.3380	0.0000	0.0000	0.0000	0.0000	NaN	NaN	NaN	2314.0000	0.0000
Wine	OPTICS (min_samples=5)	5	0.4101	0.4392	0.0358	0.1621	0.1680	0.1949	-0.1336	5.0571	1.6194	2071.7534	242.2466

Partition-based Clustering

K-Means (best-of-10 random initialization)

Accuracy: 0.972 (best overall)

Rand score: 0.962, Adjusted Rand: 0.915

SSE: 1278.76, SSB: 1035.24

Interpretation: K-Means effectively separates all three classes, with high cohesion and separation. Random initialization performs well due to well-separated centroids.

K-Means++

Accuracy: 0.966

Interpretation: Stable initialization ensures consistent cluster assignment with slightly lower accuracy than standard K-Means.

Bisecting K-Means

Accuracy: 0.837

Interpretation: Recursive splitting may create unbalanced clusters in high-dimensional space.

K-Medoids / PAM

Accuracy: 0.910

Interpretation: Medoid-based clustering is robust to noise and outliers but slightly underperforms K-Means in high-dimensional feature space.

Hierarchical Clustering (Dendrogram)

Best linkage: Ward, Accuracy: 0.927

Average, Complete, Single linkages performed poorly (0.376–0.837)

Interpretation: Ward linkage minimizes intra-cluster variance, providing clear separation between wine classes.

Density-based Clustering

DBSCAN

Accuracy: 0.399, single cluster

Interpretation: High-dimensional feature space reduces density contrast; DBSCAN fails to separate classes.

OPTICS

Accuracy: 0.410, clusters identified: [-1, 0, 1, 2, 3, 4]

Interpretation: Subcluster detection occurs but does not match true class labels effectively due to dimensionality.

Conclusion

a. Best clustering method: K-Means (best-of-10 random, accuracy 0.972)
b. Hierarchical clustering with Ward linkage is also highly effective (accuracy 0.927)
c. Density-based methods struggle due to high dimensionality.
d. Overall, K-Means and hierarchical clustering are most suitable for the Wine dataset, achieving accuracies above 90%.

Comparative Insights

```
Saved combined results to clustering_full_results.csv

Iris: best algorithm = KMedoids/PAM (accuracy=0.8467)
  -> Achieved accuracy >= 80%

Wine: best algorithm = KMeans (best-of-10 random) (accuracy=0.9719)
  -> Achieved accuracy >= 80%
```

1. **Partition-based vs Hierarchical vs Density-based Clustering**

    - **Partition-based methods** (K-Means, K-Means++, Bisecting K-Means, K-Medoids/PAM) consistently achieved the highest accuracies on both datasets.

  - **Iris:** K-Medoids/PAM was the most effective (accuracy 0.847), slightly outperforming K-Means variants.
  - **Wine:** K-Means (best-of-10 random) excelled (accuracy 0.972), demonstrating superior handling of high-dimensional continuous features.

- **Hierarchical clustering** performed moderately well, with Ward linkage being the best choice for both datasets. It achieves slightly lower accuracy than the best partition-based method but offers valuable dendrogram visualizations for cluster structure analysis.
- **Density-based methods** (DBSCAN, OPTICS) underperformed for both datasets. They struggled with overlapping clusters in Iris and high-dimensional spaces in Wine, producing either noisy clusters or fragmented groupings.

2. **Effect of Initialization (K-Means vs K-Means++)**
   - K-Means++ showed similar performance to standard K-Means but provides more stable convergence. Differences were minor for these datasets, indicating that multiple random initializations in K-Means were sufficient.
3. **Bisecting K-Means vs Standard K-Means**
   - Bisecting K-Means produced comparable results on Iris (accuracy 0.833) but slightly lower on Wine (accuracy 0.837), suggesting that recursive splitting may not capture optimal centroids in higher dimensions.
4. **Cohesion and Separation (SSE & SSB)**
   - Partition-based methods achieved lower SSE (higher cohesion) and higher SSB (better separation), especially K-Medoids/PAM for Iris

and K-Means for Wine.

- Hierarchical methods with Ward linkage also maintained good SSE/SSB balance.

- Density-based methods had poor SSE/SSB, reflecting misaligned clusters with the true class labels.

5. **Dataset Characteristics Influence Performance**

    - **Iris (4 features, moderate overlap):** Overlapping Versicolor and Virginica reduced K-Means accuracy slightly; medoid-based clustering handled overlaps better.

    - **Wine (13 features, well-separated classes):** K-Means excelled due to clearly separable clusters in a higher-dimensional space.

6. **Practical Recommendation**

    - For **low-dimensional datasets with some overlap** (like Iris), K-Medoids/PAM or K-Means are recommended.

    - For **high-dimensional, well-separated datasets** (like Wine), K-Means with multiple initializations or Hierarchical Ward linkage is preferred.

    - **Density-based clustering** should be applied only when clusters vary significantly in density or have non-convex shapes; otherwise, partition-based methods are superior.

Overall Comparison

Across both datasets, partition-based clustering algorithms consistently outperformed hierarchical and density-based methods. For the Iris dataset, K-Medoids/PAM achieved the highest accuracy, highlighting its robustness to overlapping clusters and outliers, while K-Means and K-Means++ performed comparably but slightly lower. Bisecting K-Means and Hierarchical clustering with

Ward linkage also produced reasonably high accuracy, though other linkage methods such as Average, Complete, and Single were less effective. Density-based methods, DBSCAN and OPTICS, struggled with the moderate dimensionality and compactness of the Iris clusters, producing lower accuracies and fragmented clusters.

For the Wine dataset, K-Means (best-of-N random) achieved near-perfect clustering, reflecting the well-separated and high-dimensional nature of the data. Hierarchical clustering with Ward linkage performed well but slightly below K-Means, whereas K-Medoids/PAM, Bisecting K-Means, and other hierarchical linkages showed lower accuracies. Density-based algorithms failed to form meaningful clusters due to the dataset's structure.

Overall, partition-based methods demonstrated superior cohesion and separation, evidenced by low SSE and high SSB scores, confirming effective clustering. Hierarchical clustering can provide useful structural insights via dendrograms but is sensitive to the choice of linkage, while density-based methods are better suited for datasets with irregular cluster shapes or varying densities. These results indicate that for moderate-dimensional, well-structured datasets, partition-based algorithms remain the most reliable for accurate clustering.